Lecture 6, Feb 02, 2026

6.5. Exponential Growth and Decay.

Goal. Let $y(t)$ is the value of a quantity y at time t and if the rate of change of y with respect to t is proportional to its size $y(t)$ at any time, then

$$\frac{dy}{dt} = ky \qquad (\star).$$

where k is a constant. The equation $(\star)$ sometimes called the law of natural growth ($k>0$) or the law of natural decay ($k<0$).

Thm. The only solution of the differential equation $dy/dt = ky$ are the exponential functions

$$y(t) = y(0)e^{kt}.$$

Pf We know,

$$\frac{dy}{dt} = k\,y$$

$$dy = k\,y\,dt.$$

$$\frac{dy}{y} = k\,dt.$$

$$\int \frac{1}{y}\,dy = \int \frac{dy}{y} = \int k\,dt. \qquad \frac{d}{dx}(\ln(x)) = \frac{1}{x}$$

$$\ln(y) = kt + C.$$

$$e^{\ln(y)} = e^{kt+C}$$

$$y(t) = e^{C}\cdot e^{kt}.$$

let $t=0$ then $y(0) = e^{C}\cdot e^{k(0)} = e^{C}e^{0} = e^{C}$.

$$\boxed{y(t) = y(0)\cdot e^{kt}}$$

* Radioactive Decay.

Radioactive substances decay by spontaneously emitting radiation. If $m(t)$ is the mass and m_0 the initial mass, then the model is given $\frac{dm}{dt} = km$ where k is a negative constant.

$$m(t) = m_0 e^{kt}$$

Example. The half-life of radium-226 is 1590 years.

a). A sample of radium-226 has mass 100 mg. Find a formula for the mass of the sample that remains after t years.

b). Find the mass remaining 1000 years correct to the nearest milligram.

c). When will the mass be reduced to 30 mg?

Sln. We know that

$$m(t) = m_0 e^{kt}$$

We know that $m_0 = 100$ mg.

$$m(t) = 100 e^{kt}$$

In order to find k: we use the fact that $\underbrace{m(1590)}_{m(t)} = \frac{1}{2}(100) = 50$.

$$100 e^{1590k} = 50$$

$$e^{1590k} = \frac{50}{100} = \frac{1}{2}.$$

$$\ln\left(e^{1590k}\right) = \ln\left(\frac{1}{2}\right)$$

$$1590k = -\ln(2)$$

$$k = -\frac{\ln(2)}{1590}$$

$\ln\left(\frac{1}{2}\right) = \ln(1) - \ln(2)$

$= -\ln(2)$

then $$m(t) = 100\, e^{-\frac{\ln(2)\, t}{1590}}$$

b) The mass remaining after 1000 years is

$$m(1000) = 100\, e^{-\frac{\ln(2)\,(1000)}{1590}} \approx 65 \text{ mg}.$$

c) $30 = 100\, e^{-\frac{\ln(2)\, t}{1590}}$ Solving for t.

$$\frac{30}{100} = e^{-\frac{\ln(2)\, t}{1590}}$$

$$\ln\left(\frac{3}{10}\right) = -\frac{\ln(2)\, t}{1590}$$

$$t = -\frac{1590 \ln\left(\frac{3}{10}\right)}{\ln(2)} \approx 2762 \text{ years}.$$

* Continuously compounded interest.

Example. If \$5000 is invested at 2% interest, compounded annually then after 1 year the investment is worth $\$5000(1.02) = \$5100.00$

after 2 years it's worth

$$(\$5000 \times (1.02)) \times (1.02) = \$5000\,(1.02)^2 = \$5202.00.$$

after t years it's worth

$$\$5000\,(1.02)^t = \$5000\,(1 + \underbrace{0.02}_{2\%})^t.$$

In general, if an amount A_0 is invested at an interest rate r then after t years it's worth is

$$A_0(1+r)^t.$$

Usually, interest is compounded more frequently, say, n times per year

$$A = A_0\left(1 + \frac{r}{n}\right)^{nt}.$$

For instance, after 3 years at 2% interest a \$5000 investment will be worth.

$\$5000\,(1.02)^3 = \$5306.04$ with annual compounding.

$\$5000\left(1+\frac{0.02}{2}\right)^{2(3)} = \$5307.60$ with semiannual compounding.

$\$5000\left(1+\frac{0.02}{4}\right)^{4(3)} = \$5308.39$ with quarterly compounding

$\$5000\left(1+\frac{0.02}{12}\right)^{12(3)} = \$5308.92$ with monthly compounding.

You can see that the interest paid increase as the number of compunding periods (n) increases. If we let $n\to\infty$, then we will be compunding the interest continuously and the value of the investment will be

$$A(t) = \lim_{n\to\infty} A_0\left(1+\frac{r}{n}\right)^{nt}$$

$$= A_0 \lim_{n\to\infty}\left[\left(1+\frac{r}{n}\right)^{n/r}\right]^{rt}$$

$$= A_0\left[\lim_{n\to\infty}\left(1+\frac{r}{n}\right)^{n/r}\right]^{rt} \quad , \; m=\frac{n}{r} \Rightarrow \frac{r}{n}=\frac{1}{m}$$

$$= A_0\left[\lim_{m\to\infty}\left(1+\frac{1}{m}\right)^{m}\right]^{rt}$$

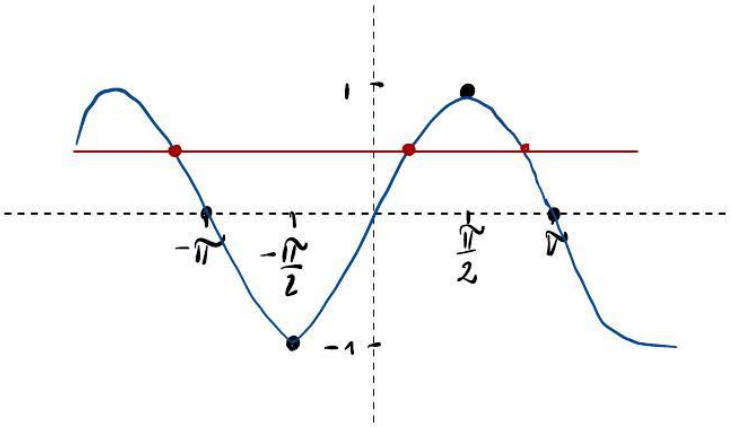
$$= A_0 e^{rt}.$$

Returning to the example of \$5000 invested for 3 years at 2% interest, we see that with continuous compounding

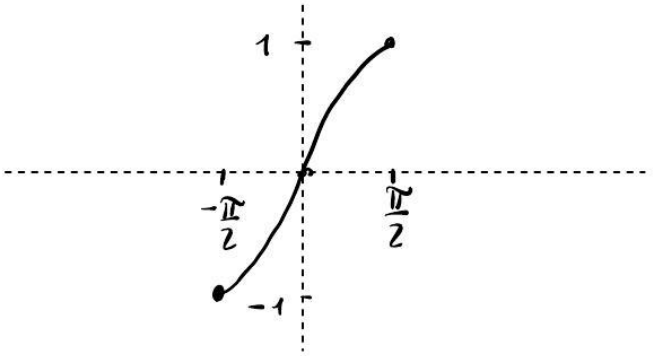
$$A(3) = \$5000\, e^{(0.02)3} = \$5309.18.$$

6.6 Inverse Trigonometric functions

let $f(x) = \sin x$.



let $y = \sin x$, $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$.



Since the definition of an inverse function says that

$$f^{-1}(x) = y \overset{\text{def}}{\iff} f(y) = x.$$

Notation:

$\sin^{-1}(x) = \arcsin(x)$

We have

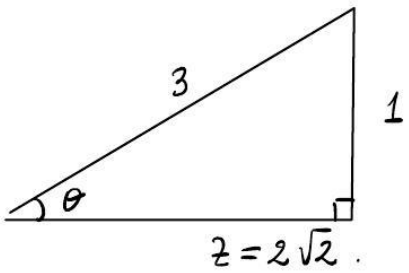
$$\sin^{-1}(x) = y \overset{\text{def}}{\iff} \sin y = x \quad \text{and} \quad -\frac{\pi}{2} \le y \le \frac{\pi}{2}.$$

Example Evaluate $\sin^{-1}\left(\frac{1}{2}\right)$, $\tan(\arcsin\frac{1}{3})$

Sln.

- $\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ b/c $\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$ and $\frac{\pi}{6} \in [-\frac{\pi}{2}, \frac{\pi}{2}]$.

- let $\theta = \arcsin\left(\frac{1}{3}\right)$ So, $\sin\theta = \frac{1}{3}$.



by the P-T

$$3^2 = 1^2 + z^2$$

$$9 - 1 = z^2 \Rightarrow z^2 = 8 \Rightarrow z = \sqrt{8} = 2\sqrt{2}.$$

$$z = 2\sqrt{2}.$$

We are looking for

$$\tan(\arcsin(\tfrac{1}{3})) = \tan(\theta) = \frac{1}{2\sqrt{2}}.$$

$$\sin^{-1}(\sin x) = x \quad \text{for} \quad -\tfrac{\pi}{2} \le x \le \tfrac{\pi}{2}$$

$$\sin(\sin^{-1}(y)) = y \quad \text{for} \quad -1 \le y \le 1.$$

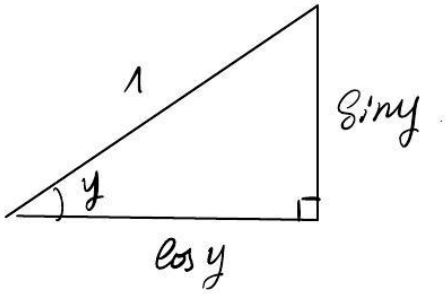
Let $y = \sin^{-1}(x)$ then $x = \sin y$. $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$.

Differentiating $\sin y = x$

$$\frac{d}{dx}(\sin y) = \frac{d}{dx}(x)$$

$$\cos y \cdot \frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{\cos y}$$



by P-T

$$\sin^2 y + \cos^2 y = 1$$

$$\cos y = \sqrt{1 - \sin^2 y}$$

$$\cos y = \sqrt{1 - x^2}$$

$$\frac{d}{dx}\left(\sin^{-1}(x)\right) = \frac{1}{\sqrt{1-x^2}} \quad -1 < x < 1$$

Example. If $f(x) = \sin^{-1}(x^2 - 1)$ find

a). Dom f.

b). $f'(x)$.

c). Dom f'

Sln. a). Since the domain of the inverse sine function is $[-1, 1]$. the domain of f is

$$\{x \mid -1 \le x^2 - 1 \le 1\} = \{x \mid 0 \le x^2 \le 2\}$$

$$= \{x \mid 0 \le |x| \le \sqrt{2}\}.$$

$$= [-\sqrt{2}, \sqrt{2}]$$

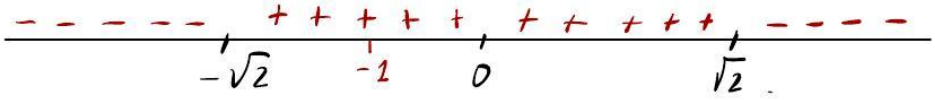
b). $f'(x) = \frac{1}{\sqrt{1-(x^2-1)^2}} \cdot \frac{d}{dx}(x^2-1)$. by C-R.

$$= \frac{2x}{\sqrt{2x^2 - x^4}}.$$

c). Solving $2x^2 - x^4 > 0$

$$x^2(2 - x^2) > 0$$

$$x^2(\sqrt{2} - x)(\sqrt{2} + x) > 0$$



Dom f' = $(-\sqrt{2}, 0) \cup (0, \sqrt{2})$.